1.) PCA : $X \longrightarrow \dfrac{X-\mu}{\sigma}$

Allcott & Kenniston (2018)

"1 std dev above the mean"

— When units are hard to define

2.) $$\begin{bmatrix} Y_{1t} \\ Y_{2t} \end{bmatrix} = \begin{bmatrix} \phi_{11} & 0 \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} Y_{1t-1} \\ Y_{2t-1} \end{bmatrix} + \begin{bmatrix} e_{1t} \\ e_{2t} \end{bmatrix}$$

ECON 582 Quiz Section 3

Tyson Ramirez

[1]Department of Economics
University of Washington

Spring 2026

Identification

- What IS identification? What is meant by the term?
- Any relation to math we're familiar with?
- Identification vs. Prediction?
- See Lewbel (2019) for more discussion re: identification

Zotero

def: "<u>Identification</u>"

Model parameters or features being <u>uniquely</u> determined from the observable pop. that generates the data

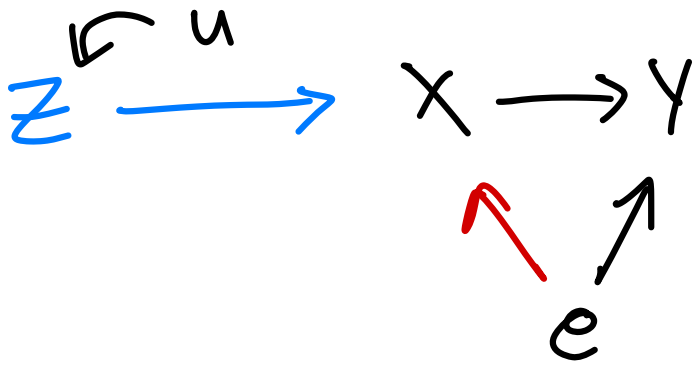
↑ Point Identification

- Systems of (Linear) Eqns:

Eqns > # Unknowns : No Sol'n | Under determined
Eqns = # Unknowns : Unique Sol'n / Just Determined
Eqns < # Unknowns : Many Sol'ns / Over Determined

Instrumental Variables 1

- Causal Graph (+ ex.)



ex: Card (1995)

X = educ, Y = salary

e = ability, Z = grew up near a university

ex: Kanzig (2021)

X = Oil Prices Y = GDP

e = Global AD, Z = OPEC Announcements

Instrumental Variables 1

- Causal Graph (+ ex.)
- Structural Model for IV:
  Suppose we want to estimate the model

Y_z

$$Y = \beta_0 + X_1\beta_1 + X_2\beta_2 + e$$

... but $E(Xe) \neq 0$ (specifically $E(X_2 e) \neq 0$)
... BUT $\exists$ some Z s.t. $E(Ze) = 0$

⋆ We could make it work using *only* these moment conditions, but imposing more (correct) structure yields better information!

Instrumental Variables 2

- More structure:

$$Y = \beta_0 + X_1\beta_1 + X_2\beta_2 + e, \quad E(Xe) \neq 0$$
$$X_2 = \Gamma_0 + X_1\Gamma_1 + z\Gamma_2 + u, \quad E(Ze) = E(Zu) = 0$$

where $Z = \begin{pmatrix} X_1 \\ z \end{pmatrix} = \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix}$

$\dim(Z) = \dim(Z_1) + \dim(Z_2)$
$= \ell = \cancel{k}\ 3$
$\dim(X) = \dim(X_1) + \dim(X_2)$
$= k = \cancel{\ell} = 3$

- Necessary conditions:
  1. Relevance: $E(X_2 z) \neq 0$
  2. Excludability: $E(Ze) = 0$

ECON 587

Instrumental Variables 3

- Navigating things:

$$Y_i = X_i'\beta + e_i, \quad E(Xe) \neq 0$$
$$X_i = \Gamma' Z_i + u_i, \quad E(Ze) = E(Zu) = 0$$

- ILS, IV, 2SLS???

In matrix form:

$$Y_i = X_i'\beta + e_i$$
$$\underset{K\times 1}{X_i} = \underset{K\times l}{\Gamma'}\underset{l\times 1}{Z_i} + \underset{K\times 1}{U_i}$$

$$Y = (\Gamma' Z_i + U_i)'\beta + e_i$$
$$= (Z_i'\Gamma + U_i')\beta + e_i$$
$$= Z_i'\Gamma\beta + U_i'\beta + e_i$$

We have

$$V_i = U_i'\beta + e_i$$
$$\lambda = \Gamma'\beta$$

We want β

$$\lambda = \Gamma'\beta$$
$$\beta = \Gamma^{-1}\lambda$$

Γ is $l\times K$ & $l = K$ & rk'd

$$\hat{\beta}_{FZS} = \hat{\Gamma}^{-1}\hat{\lambda} \xrightarrow{p} \beta$$

$$\hat{\Gamma} = (Z'Z)^{-1}Z'X \longrightarrow \Gamma$$
$$\hat{\lambda} = (Z'Z)^{-1}Z'Y \longrightarrow \lambda$$

IV: $E(Z\varepsilon) = 0$

$$E(Z(Y - X'\beta)) = 0$$

$\vdots$

$$\beta = E(Z'X)^{-1}E(Z'Y)$$
$$\hat{\beta}_{IV} = (Z'X)^{-1}(Z'Y)$$

$\rightarrow$ still use $l = k$ for above to work

* When $l > k$? "Over identified"

Recall $Y_i = Z_i'\lambda + V_i$

where $\lambda = \Gamma\beta$

$X_i = \Gamma' Z_i + U_i$

$$Y_i = Z_i'\Gamma\beta + V_i$$

def: $W_i = Z_i\Gamma$ $\longrightarrow$ $Y_i = W_i\beta + V_i$

$$\hat{\beta}_{2SLS} = (W'W)^{-1}W'Y$$

$$= ((Z'\Gamma)'(Z'\Gamma))^{-1}(Z'\Gamma)'Y$$

$$= (\hat{\Gamma}'ZZ'\hat{\Gamma})^{-1}\hat{\Gamma}'ZY = \hat{\beta}_{2SLS}$$

$$\hat{\Gamma} = (Z'Z)^{-1}Z'X$$

Instrumental Variables 4

- End of the day:
  Predict X_2 using Z, use fitted X_2 in original model

$$Y = \beta_0 + X_1\beta_1 + \hat{X}_2\beta_2 + e$$

- Properties (of 2SLS)?
  - 2SLS is unbiased
  - 2SLS is consistent
  - 2SLS is asymptotically Normal
  - Under homoskedasticity:
    $Var(\hat{\beta}_{2SLS}) = \sigma^2 \left(E(X'Z)E(Z'Z)^{-1}E(X'Z)\right)^{-1}$
    and it is efficient
  - Weak instruments complicates all of these! $(E(XZ) \to 0)$

* What is "Strong"?
in 1st Stage: $F > 10$
$\vdots$
$F > 16$

* $\hat{\beta}_2$ = LATE

Preview: GMM

- Next topic: Generalized Method of Moments (GMM)
- Main idea: We can estimate model parameters using ONLY moment conditions*
- Relaxes strong distributional assumptions and allows both linear and non-linear models
- GMM nests ALL of the estimators you've likely seen so far + more!